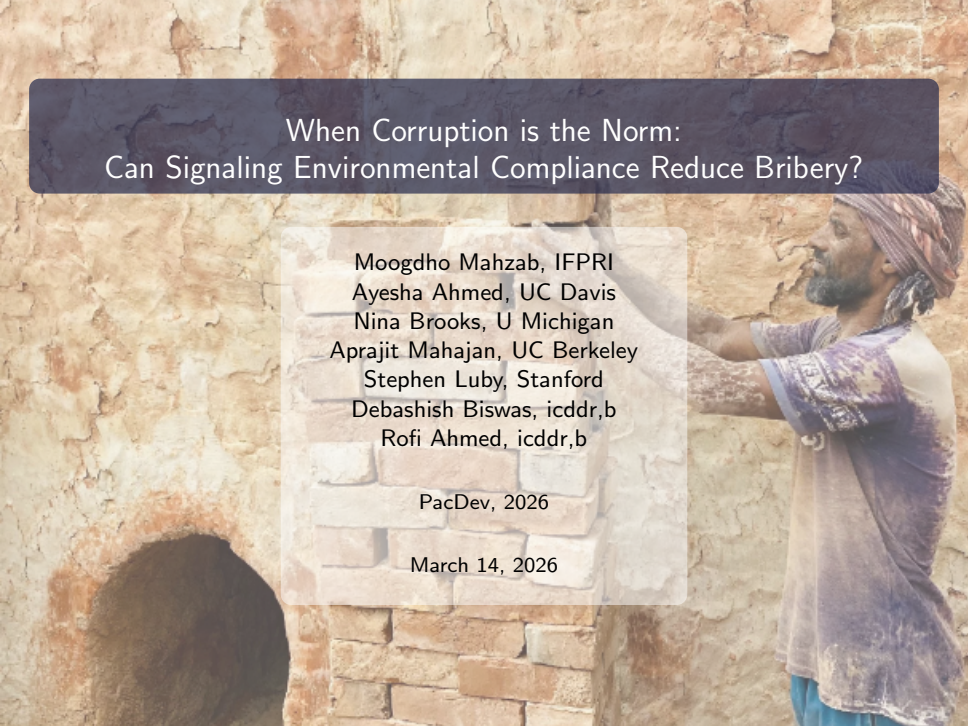


A man with a beard, wearing a purple patterned shirt and a turban, is shown from the side, working on a brick wall. He is holding a brick in his right hand, about to place it. The wall is made of reddish-brown bricks and has a rough, textured appearance. In the background, there is a large, dark, arched opening in the wall.

When Corruption is the Norm: Can Signaling Environmental Compliance Reduce Bribery?

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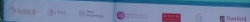
PacDev, 2026

March 14, 2026

Introduction

এই ইটভাটা একটি গবেষণা প্রকল্পের অন্তর্ভুক্ত

প্রকল্পের নাম	: নির্দল বায়ু এবং সুবাসের জন্য উন্নত পদ্ধতিতে ইট উৎপাদন প্রক্রিয়া সম্প্রসারণ প্রকল্প
ইটভাটার নাম	: মেসার্স রনি ব্রিক্স
ইটভাটা মালিকের নাম	: মোঃ মুনসুর আলী
ঠিকানা	: মোহাম্মাদালা, ভাকুর্তা, শাহার, ঢাকা।
প্রকল্পের মেয়াদ	: অক্টোবর ২০২০ - জুন ২০২৫
প্রকল্প বাস্তবায়নে	: আইসিটিভিআর,বি, বাংলাদেশ প্রকৌশল বিদ্যাবিদ্যালয়, গ্রীনটেক নলেজ সলিউশন প্রাই ভিমেটেক, বোস্টন বিশ্ববিদ্যালয়, এবং স্ট্যানফোর্ড বিশ্ববিদ্যালয়



Motivation and Concept

- Weak regulation sustains corruption in informal industries, where firms rely on illicit payments to operate under non-compliance.
- Frontline bureaucratic discretion limits the effectiveness formal enforcement.
- Compliance alone is insufficient when regulatory clearance depends on subjective judgment.
- Firms may strategically reshape interactions with officials through credible signals of compliance. .
- Brick kilns in Bangladesh provide a canonical setting of informality, violation (environmental), and discretionary enforcement.

The Signaling Tradeoff

① Rent Extraction Hypothesis:

- Signals might increase bribe demands if officials perceive compliant kilns being more profitable.

② Bureaucratic Preference Hypothesis:

- Signals might reduce demands if officials genuinely value environmental improvements or fear oversight.

Research Question: Can firms reduce illicit costs by signaling regulatory compliance?

Contributions to the Literature

- **Corruption & Firm Behavior:** Extends the literature on corruption and firm behavior by showing that credible compliance signals can deter rent extraction in high-discretion environments (Banerjee et al., 2012; Olken and Pande, 2012; Svensson, 2003).
- **Enforcement & Monitoring:** Complements work on enforcement and monitoring by highlighting a bottom-up channel that operates even under weak regulatory capacity (Amin and Soh, 2021; Ha et al., 2024).
- **Measurement:** Contributes to the measurement of corruption in informal sectors using detailed data on bribes, compliance fees, and political donations (Fisman and Svensson, 2007; Hanna and Wang, 2017).

Background



এই ইটভাটা একটি গবেষণা প্রকল্পের অন্তর্ভুক্ত

প্রকল্পের নাম	ঃ নির্মল বায়ু এবং সুবাসের জন্য উন্নত পদ্ধতিতে ইট উৎপাদন প্রক্রিয়া সম্প্রসারণ প্রকল্প
ইটভাটার নাম	ঃ মেসার্স মেম্বা ব্রিকস (BMB)
ইটভাটা মালিকের নাম	ঃ মোঃ ভোকাবুল হোসেন বকুল
ঠিকানা	ঃ কাচারিপাড়া, পিরুজাদী, ভাওয়াল মির্জাপুর, গাজীপুর সদর, গাজীপুর।
প্রকল্পের মেয়াদ	ঃ অক্টোবর ২০২৩ - জুন ২০২৫
প্রকল্প বাস্তবায়নে	ঃ আইসিটিভিআর, লি, বাংলাদেশ প্রকৌশল বিশ্ববিদ্যালয়, গ্রীনটেক নলেজ সলিউশন প্রাইভেট লিমিটেড, বোস্টন বিশ্ববিদ্যালয়, এবং স্ট্যানফোর্ড বিশ্ববিদ্যালয়



Kiln Siting Regulations and Patterns of Non-Compliance

The Brick Manufacturing and Kiln Establishment (Control) Act, 2013 .pdf

(2) After the enforcement of this Act, the Department of Environment, or any other authority under any law shall not, give permit or clearance certificate or license, by whatever name called, for establishing brick kilns within the prohibited areas.

(3) ¹⁷***] No person shall establish brick kilns in the following distances or places, namely:-

- (a) within minimum 1 (one) kilometer distance, from the boundaries of the prohibited areas;
- (b) ¹⁸***] within 2 (two) kilometers distance from the boundaries of public forest;
- (c) in case of the establishment of brick kiln on the top or slope or the surrounding ground surface of any hill or hillock, within minimum ½ (half) kilometer distance from the foot of the said hill or hillock;
- (d) in case of the establishment of brick kilns in the hill districts, in any place other than the place determined by the hill districts Environment Development Committee;
- (e) within minimum 1 (one) kilometer distance, from any special structure, railways, educational institutions, hospitals and clinics, research institutions, or any other similar place or institution ¹⁹[:]

²⁰***]

Conceptual Framework: Intuition and Notation

Model Setup:

Kilns are randomly assigned to adopt Zigzag 2.0 and display a public banner, which serves as a credible, public signal of regulatory compliance and potentially higher profitability. Government officers are heterogeneous in their preferences, differing in the weight they place on rent extraction versus environmental improvement.

Officer's Decision: The officer chooses a bribe level b when inspecting a kiln.

$$\max_b U_O = (1 - \beta)(b + \lambda\Delta E) + \beta\gamma E - \varepsilon b$$

- b : bribe demanded
- $E \in \{0, 1\}$: banner displayed
- β : weight on environmental motives
- Δ : extra profit from Zigzag 2.0
- λ : weight on rent extraction from profits
- γ : utility from pollution reduction
- ε : expected penalty

Conceptual Framework: Predictions

First-order Condition:

$$\frac{\partial U_O}{\partial b} = (1 - \beta) - \varepsilon = 0 \implies \beta^* = 1 - \varepsilon$$

Empirical Predictions:

- **Low- β officers:** banners may raise bribes by signaling higher profits.
- **High- β officers:** banners reduce bribes by reinforcing environmental compliance.
- With persistent signals, the environmental channel becomes more salient over time.

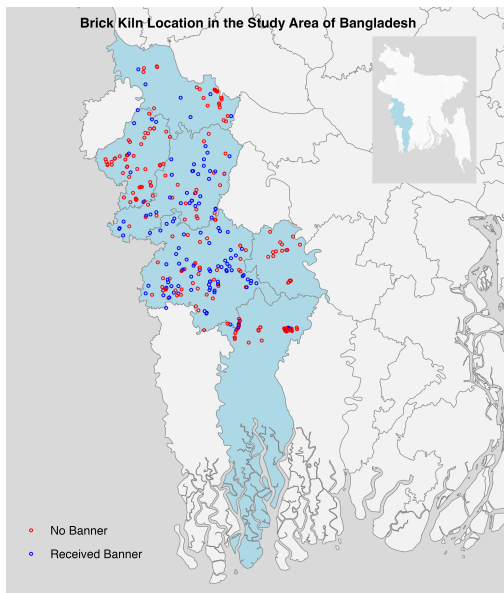
Study Design

এই ইটভাটা একটি গবেষণা প্রকল্পের অন্তর্ভুক্ত

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ইটভাটার নাম	: স্বপন অটো ব্রিকস ইন্ডাস্ট্রিজ লিঃ
ইটভাটা মালিকের নাম	: মোঃ ওবায়দুল্লাহ
ঠিকানা	: পাইনশাইল চক, ভাওয়াল মির্জাপুর, গাজীপুর সদর, গাজীপুর।
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Study Site

- RCT during the 2022–23 brick-firing season across Jessore, Khulna, Jhenaidah, Chuadanga, Kushtia, and Narail in Khulna Division, Bangladesh.
- Randomized kilns received technical guidance to make brick making environmentally efficient (Brooks et al. 2025: *Science*).
- Fulfilling compliance to the technological adoption entitled kilns to receive banners.



Study Design and Data

- Banners awarded only to treated kilns that verified adoption of cleaner practices (stacking + single fireman).
- Banners serve as credible, public compliance signals endorsed by reputed institutions.
- Analysis sample: 336 kilns; 132 received banner and 204 did not; banner display duration varies from 3 to 34 weeks.
- Outcomes from baseline and endline surveys: bribes, compliance costs, and political donations (per 1,000 bricks).
- Kilns were balanced in illicit payments behavior at the baseline.
- Qualitative data: interview with government officials.
- Geo-spatial kiln characteristics information.

This brick kiln is under a performance improvement project to reduce air pollution

Project title: Transforming brick manufacturing to promote clean air and better health in Bangladesh

Name of brick kiln: Shohagh bricks

Owner of the kiln: Mr. Abul Kashem

Address: Monirampur, Jashore

Project duration: October 2021 to June 2022

Project conducted by: icddr,b, BUET, Greentech Knowledge Solutions, Stanford University



Banners in front of the Kilns



Identification Strategy

- Banner display is endogenous: kilns that adopt cleaner practices receive banners.
- Instrument: randomized assignment to environmental treatment arms (Z_{is}).

$$\text{First stage: } B_{is} = \theta_0 + \theta_1 Z_{is} + \gamma_s + X'_{is} \Gamma + \varepsilon_{is}$$

$$\text{Second stage: } Y_{is} = \beta_0 + \beta_1 \hat{B}_{is} + \gamma_s + X'_{is} \Gamma + u_{is}$$

- B_{is} : banner duration (weeks) or receipt indicator.
- Y_{is} : illicit payments per 1,000 bricks.
- Controls: subdistrict fixed effects and baseline kiln characteristics.
- Inference: standard errors clustered at intervention level.

→ β_1 identifies the LATE of compliance signaling for banner recipient kilns.

Results



Summary Statistics of Illicit Costs

Table: Proportion of Kilns Making Illicit Payments

Payments Type	Total Kiln	N (%)
Bribes	336	258 (76.8)
Compliance Cost	336	336 (100.0)
Political Donations	336	316 (94.0)

Table: Illicit Costs at Baseline and Endline (N = 338)

Cost Type	Baseline (2022)		Endline (2023)	
	Per Season	Per 1,000 Bricks	Per Season	Per 1,000 Bricks
Bribes	202,391	46	144,996	27
Compliance Cost	658,576	155	546,811	101
Political Donations	148,815	32	70,658	13
Illicit Share of Total Costs	3%	3.2%	1.9%	2%

► Balance Table

Association of kiln characteristics with illicit costs

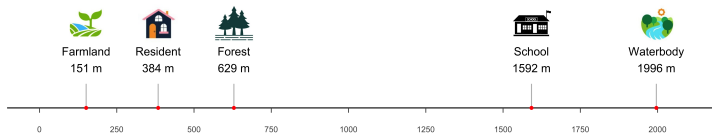
Table: Association Between Individual Kiln Characteristics and Illicit Payments

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Pol. Donations</i>
School Distance	0.002 (0.002)	0.001 (0.005)	0.002 (0.002)
Waterbody Distance	-0.000 (0.002)	0.001 (0.005)	0.001 (0.001)
Forest/Garden Distance	-0.006** (0.003)	-0.012* (0.006)	0.005 (0.004)
Residential Area Distance	-0.005 (0.006)	-0.002 (0.012)	0.000 (0.005)
Farmland Distance	0.001 (0.002)	0.008 (0.008)	0.001 (0.001)
Plants and Trees (%)	0.752*** (0.274)	-0.621 (0.380)	0.146 (0.140)
Violation Index	-0.859 (1.913)	-1.249 (3.308)	2.425 (1.868)

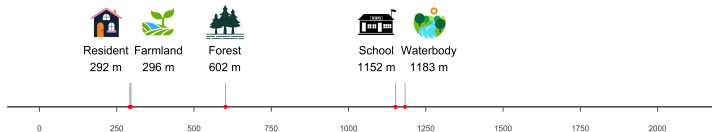
Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Kiln Siting Regulations and Patterns of Non-Compliance

Average distances from kilns to regulated areas



(a) All kilns



(b) Demolished kilns

Result: Effect of Ever Receiving a Banner

Table: IV Estimates of Banner Receipt on Illicit Payments per 1,000 Bricks

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Banner Received = 1	-5.706** (2.279)	-5.560* (3.232)	-0.839 (1.048)
Constant	40.882*** (5.473)	142.516*** (13.394)	12.852*** (1.974)
Observations	315	315	315
R-squared	0.193	0.235	0.211
Control Mean	44.9	152.8	34.6

Notes: 2SLS estimates. Instrument: Randomized Environmental Intervention Assignment. All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

► First Stage

Result: Effect of Weekly Banner Display

Table: IV Estimates of Weekly Banner Display on Illicit Payments per 1,000 Bricks

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
Banner Duration (Weeks)	-0.294** (0.126)	-0.312** (0.157)	-0.051 (0.054)
Constant	39.847*** (5.059)	141.949*** (13.680)	12.798*** (2.029)
Observations	315	315	315
R-squared	0.193	0.235	0.211
Weekly Cost [†]	2.4	8.1	1.8

Notes: 2SLS estimates. Instrument: Randomized Environmental Intervention Assignment. All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. [†] Weekly Cost is the baseline control mean scaled by the average banner duration among treated kilns (18.9 weeks). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Robustness Check: Controlling for Production Scale

Table: IV Estimates of Banner Duration on Illicit Payments (Controlling for Production)

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A</i>			
Banner Duration (Weeks)	-0.216** (0.108)	0.013 (0.173)	-0.027 (0.060)
Annual Production (Lakh)	-0.515*** (0.124)	-1.967*** (0.121)	-0.172*** (0.030)
Constant	65.566*** (1.087)	246.227*** (12.409)	21.313*** (1.279)
<i>Panel B</i>			
Banner Received = 1	-4.205** (1.928)	0.465 (3.342)	-0.414 (1.128)
Annual Production (Lakh)	-0.529*** (0.123)	-1.968*** (0.121)	-0.174*** (0.028)
Constant	67.092*** (0.618)	246.040*** (12.713)	21.441*** (1.485)
Observations	315	315	315

Note: Models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs.

Robustness Check: Signaling Effect on the Non-Adopters

Table: Effect of Banner Duration on Illicit Payments for Technology Non-adopters

	<i>Bribe Payments</i>	<i>Compliance Costs</i>	<i>Political Donations</i>
<i>Panel A</i>			
Banner Duration (Weeks)	-1.954*** (0.481)	-0.864 (1.631)	-0.463* (0.264)
Constant	114.726***	181.608***	28.144***
<i>Panel B</i>			
Banner Received = 1	-35.658*** (8.926)	-13.025 (30.851)	-8.001* (4.748)
Constant	122.114***	182.139***	29.415***
Observations (Panel A/B)	152	152	152

Note: Models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs.

Insights from Government Interviews

- Air pollution drives enforcement (fixed chimney, black smoke is the main trigger).
- Multiple violations matter (technology, siting, land, fuel use).
- Banned kilns still operate despite formal prohibitions.
- Enforcement is constrained by resources and political pressure.
- Intervention acted as a signal of cleaner and also profitable production.

Discussion

এস, এ, ব্রিকস

উন্নতমানের ইট প্রস্তুতকারক ও পরিবেশক,
গেমবাগ, জাহাঙ্গীরনগর, যশোর।

SAB

মেসার্স ব্রাদার্স ট্রেডিং কোং

বি.সি.আই.সি. অনুমোদিত সার ডিলার
১নং গেমবাগ ইউনিয়ন

সারের মূল্য তালিকা

ইউরিজ - ১৩৫০/- অণু, ড.পি - ১৩৫০/-
ডি.এ.পি - ১০৫০/- টি.এস.পি - ১০৫০/-

ফিউচারি সারের মূল্য তালিকা

এই ইটভাটা একটি গবেষণা প্রকল্পের অন্তর্ভুক্ত

প্রকল্পের নাম	ঃ নির্মল বাঘ এবং সুখাছুরের জন্য উন্নত পদ্ধতিতে ইট উৎপাদন প্রক্রিয়া সম্প্রসারণ প্রকল্প
ইটভাটার নাম	ঃ এস, এ, ব্রিকস
ইটভাটা মালিকের নাম	ঃ মোঃ শাহজাদুল হোসেন
ঠিকানা	ঃ গেমবাগ, অভয়নগর, যশোর।
প্রকল্পের মেয়াদ	ঃ অক্টোবর ২০২৩ - জুন ২০২৫
প্রকল্প বাস্তবায়নে	ঃ আইসিডিডিআর, বি, বাংলাদেশ প্রকৌশল বিশ্ববিদ্যালয়, গ্রীন্ডটেক নশেজ সলিউশন গ্রাঃ লিমিটেড, বোস্টন বিশ্ববিদ্যালয়, এবং স্ট্যানফোর্ড বিশ্ববিদ্যালয়

icddr,b

IFPRI

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Discussion

- Persistent compliance signals reduce bribes and illicit compliance costs in a weakly regulated, informal industry.
- Compliance alone is insufficient, credibility of the signal is essential to altering rent seeking behavior.
- Signaling complements technology adoption: rather than triggering higher rent extraction, it amplifies productivity and environmental gains.
- Results could be generalized to other informal, high-discretion sectors where enforcement is weak and corruption is entrenched.
- Finally, low-cost, bottom-up compliance signals can mitigate corruption and environmental harm even without strong institutions.

A man with a beard, wearing a purple patterned headscarf and a purple t-shirt, is shown from the side, stacking a tall column of red bricks. He is standing in front of a wall made of rough, reddish-brown mud or plaster. To the left of the brick column is a dark, arched opening in the wall. The text "Thank you!" is overlaid in white on the brick column.

Thank you!

References I

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Appendix: First Stage

Table: Banner Duration (Weeks) on Environmental Intervention Assignment.

	<i>Banner Duration (Bribes)</i>	<i>Banner Duration (Compliance)</i>	<i>Banner Duration (Pol. Donations)</i>
Env Intervention = Technical	11.318*** (0.337)	11.219*** (0.397)	11.257*** (0.484)
Env Intervention = Incentive	10.618*** (0.421)	10.817*** (0.501)	10.787*** (0.600)
Constant	4.321 (2.428)	8.024 (4.181)	4.212 (2.525)
Observations	315	315	315
R-squared	0.561	0.574	0.561

Notes: All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[▶ Back](#)

Appendix: First Stage

Table: Banner Receipt on Environmental Intervention Assignment.

	<i>Banner Received (Bribes)</i>	<i>Banner Received (Compliance)</i>	<i>Banner Received (Pol. Donations)</i>
Env Intervention = Technical	0.584*** (0.018)	0.581*** (0.020)	0.585*** (0.021)
Env Intervention = Incentive	0.579*** (0.020)	0.585*** (0.020)	0.586*** (0.024)
Constant	0.369 (0.208)	0.456 (0.261)	0.353 (0.197)
Observations	315	315	315
R-squared	0.589	0.592	0.588

Notes: All models control for subdistrict fixed effects, kiln characteristics, and baseline illicit costs. Standard errors clustered at the intervention unit level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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Appendix: Balance Table

Table: Balance table of Baseline Characteristics by Banner Treatment.

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Variable	Received Banner = 0		Received Banner = 1		Pairwise t-test
	N	Mean	N	Mean	P-value
Educational Institute Distance	198	1514.02 (66.42)	128	1710.95 (90.22)	0.75
Waterbody Distance	198	1845.32 (135.69)	128	2212.20 (158.61)	0.01**
Forest or Garden Distance	197	610.42 (50.56)	126	666.38 (76.97)	0.64
Residential Area Distance	197	368.82 (28.21)	127	412.24 (37.54)	0.32
Farmland Distance	194	198.28 (51.75)	126	79.41 (19.74)	0.43
Plants and Trees (%)	198	21.08 (1.22)	128	27.87 (1.55)	0.08*
Waste Management Septic Tank	198	0.01 (0.01)	128	0.01 (0.01)	0.70
Demolished Kiln	197	0.06 (0.02)	128	0.09 (0.02)	0.47
Bribes-1k br	205	44.92 (2.80)	131	49.03 (4.55)	0.38
Compliance-1k br	205	152.81 (5.68)	131	159.69 (9.83)	0.58
Pol. Donations-1k br	205	34.59 (2.33)	131	29.31 (2.48)	0.12
F-test of joint significance (P-value); N = 314		1.57	(0.11)		